Faculty of Engineering, University Of Jaffna

Department Of Computer Engineering

EC1011 – Computing

Lab 02 - Operating System Installation

Date: 30th December 2018 Duration: 3 Hours Instructions:

1. Do this lab session according to the procedures mentioned in this instruction sheet.
2. Prepare a simple report named ***Lab02_2019EXXX*** *(XXX – Your Registration Number)* which contains screen-shots of all the steps which are done by you during the os installation. Make sure you describe the step and not just attach the screen shots. Use the given template to make your report.
3. Submit your report in a ***pdf*** format via teams ***on or before 26th December 2019 at 05.30Pm***.

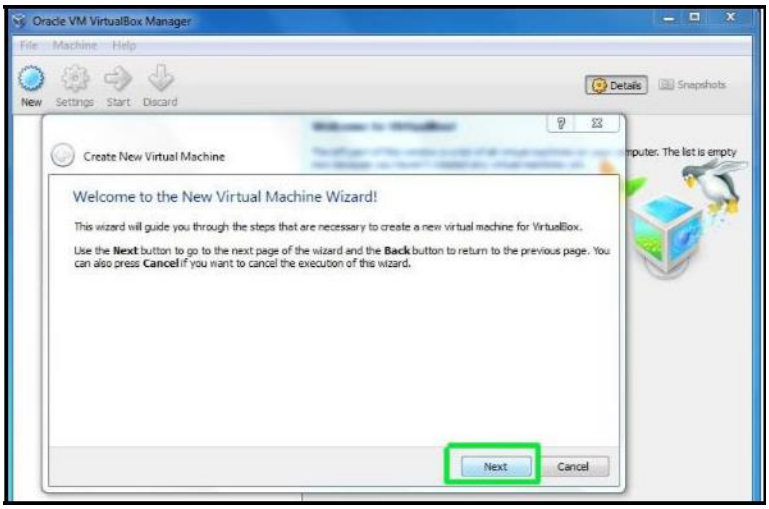
Teams → EC1011 → Assignments → Lab_02_GroupY(Y is your group)

Run Ubuntu within Windows Using VirtualBox

Create a Virtual Machine

i. First open the VirtualBox application in your PC

ii. click the New icon on the toolbar.



The first thing you need to do is define the type of virtual machine you wish to create. Enter a descriptive name into the Name box, use your registration number for the name as 2019eXXX (*XXX – Your Registration Number*).

iii. *Select Linux as the Type.*

iv. *Choose Ubuntu as the Version.*

v. *Click Next to continue.*

Note: Make sure you choose the correct version. You must choose 32-bit if your host computer is a 32-bit machine. If you are using a 64-bit machine you can choose either 32-bit or 64-bit but obviously, 64-bit is recommended. Include the screen shot of checking your PC's architecture.

Allocate Memory to The Virtual Machine

1. The next screen asks you to set how much memory you wish to give over to the virtual machine.

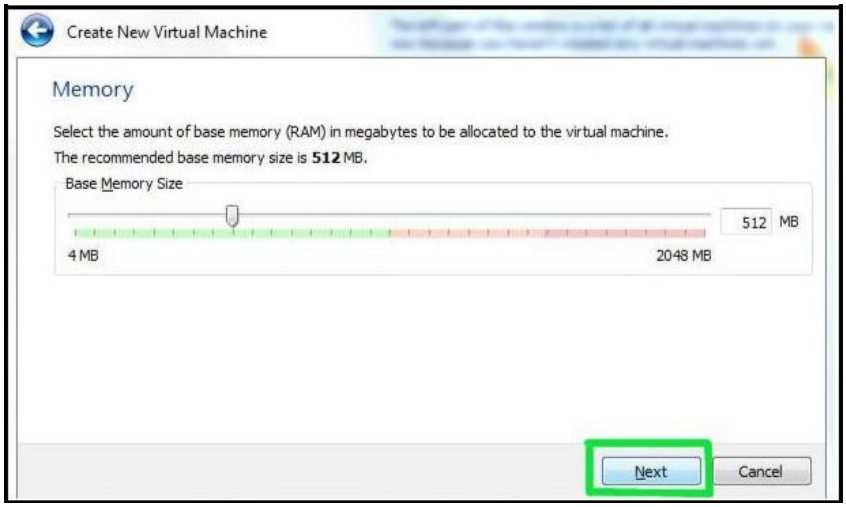
2. You should not go below the minimum specified and you should also make sure you leave enough memory for the host operating system (Windows) to keep running.

3. 4 gigabytes of RAM is recommended for Ubuntu 18.04. Further, 512 megabytes is recommended for Ubuntu 16.04 however, it is recommended to increase the bar to 2048 megabytes. Do not exceed the installed memory capacity of your PC. Include the screen shot of checking your PC's installed memory.

Create a Virtual Hard Drive

The next three steps are all about allocating disk space to the virtual machine.

If you want to run Ubuntu as a live image, then you do not need to create a hard drive at all



but for installing Ubuntu you will need to.

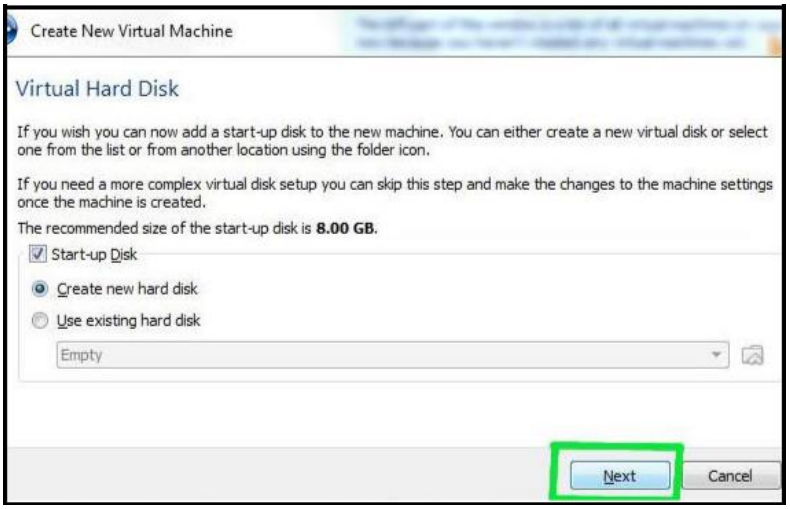
i. Select Create a virtual hard drive now.

ii. Click Create.

iii. You will be asked to select the type of hard drive to create.

iv. The default VDI file type is the one native to VirtualBox, so select VDI.

v. Click Next.



When deciding on the way the hard drive is created you can choose to opt for a fixed size hard drive or a dynamically sized hard drive.

Important: At this point, no partitioning occurs on your actual hard drive. All that happens is that a file is created on your computer which acts as the hard drive.

A fixed size disk creates the hard drive to be the maximum size you define straight away whereas a dynamically sized disk adds space to the file as it is required up to the maximum size that you specify.

A fixed size disk performs better because as you install software within the virtual machine it doesn't have to increase the file size on the fly. If you have enough disk space, then I recommend this option.

i. *Select your desired hard drive type.*
ii. *Click Next.*

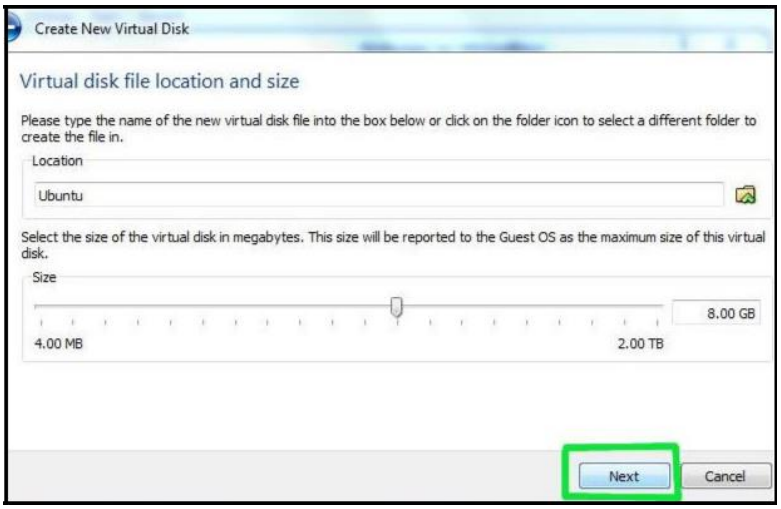
iii. *After specifying the hard drive type and the way the disk is allocated you are asked to specify how much disk space you are going to give over to the Ubuntu Virtual Machine.*

Important: *Do not go below the minimum set and create enough disk space to make it worthwhile. It is recommended to use at least 10 gigabytes.*

iv. *Choose where you wish to save the virtual machine.*

v. *Specify the disk size.*

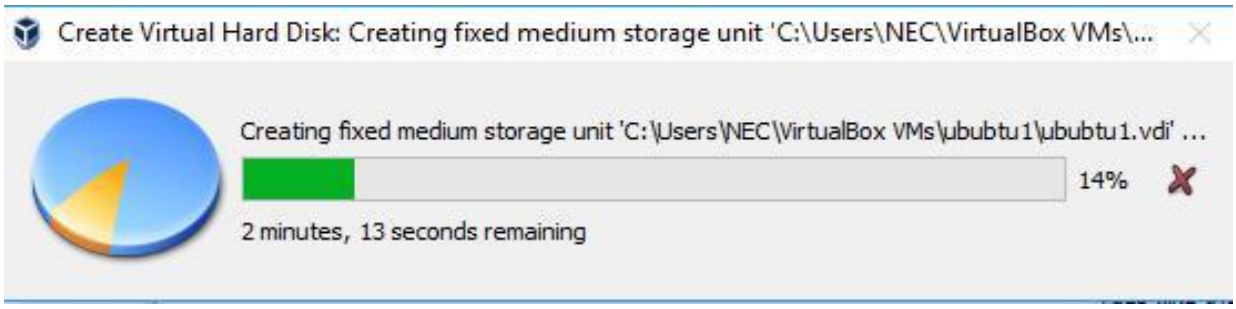
vi. *Click Next.*



vii. *Click Create.*

Recommend: *read the Displayed massages carefully.*

Note: *Creating Virtual hard Disk dialog box*



Start the Virtual Machine

The Virtual Machine has now been created and you can start it by pressing the Start button on the toolbar.

To load the Ubuntu ISO file:

i. *Go to the Network folder*

ii. *Type* \\10.222.1.6 *in the address bar*

iii. *Go the folder named "Sofware_FirstYear" then to the folder "Linux"*

iv. *Copy the iso file with the Ubuntu ISO you have chosen (use "ubuntu-18.04.3-desktop-amd64" if your pc can support a 64 bit architecture or else use "ubuntu-16.04.6-desktop-x32" if your pc supports only 32 bit architecture)*

The first boot requires you to select a startup disk.

i. *Click the folder icon.*

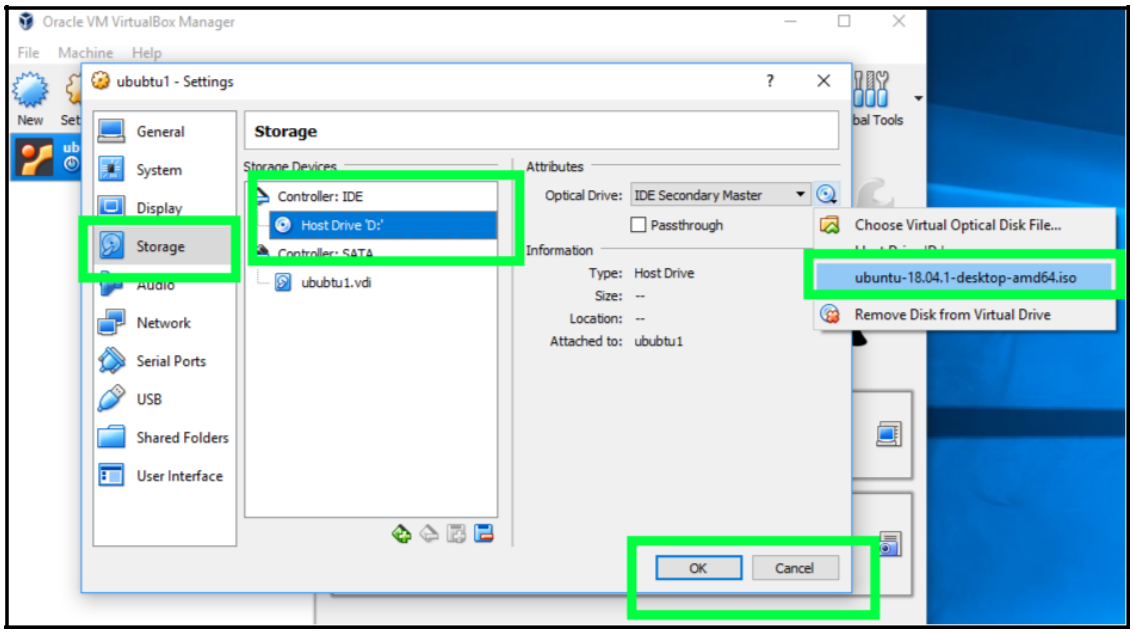
ii. *Navigate to the folder which has the Ubuntu ISO file in your computer and select Ubuntu ISO.*

iii. *Click Ok.*

Install Ubuntu Within VirtualBox

Now we are into the nitty-gritty of installing Ubuntu.

The first step is to choose the installation language.



i. Choose a language as English us.

ii. Click Continue.

iii. A screen appears showing how prepared you are for installing Ubuntu.

Recommend: *If you are using a laptop make sure your computer is either plugged in or has enough battery life. Connect to a power source especially if you plan on installing updates as you go.*

iv. There are two checkboxes at the bottom of the screen. Choose whether to install updates as you go.

v. Then choose whether to install 3rd party software.

Note: *If you have a fast-enough internet connection it is worth updating as you go but if you don't, I would recommend installing Ubuntu and updating later.*

Recommend: *Not installing the 3rd party software at this stage. This can be done post installation.*

vi. Click Continue.

Partitioning a Virtual Hard Drive

The Installation Type screen asks you how you wish to partition the hard drive.

When installing on a real hard drive this step causes people anguish. Do not panic though as this will only touch your virtual hard drive and will not affect Windows in any way whatsoever.

i. *Select Erase disk and install Ubuntu.*
ii. *Click Install Now.*
iii. *The installation begins, and the files are copied to the virtual hard drive.*

Choose Your Locations

Whilst this is going on you will be asked to choose your location. This sets the time zone for Ubuntu and makes sure the all-important clock shows the right value.

i. *Click the map to choose your location.*
ii. *Click Continue.*

Choose Your Keyboard Layout

A final couple of steps require you to choose your keyboard layout and create a user.

i. *Choose the language for your keyboard.*
ii. *Choose the type of keyboard.*
iii. *Click Continue.*

Create a User

From the Who are you screen:

i. *Enter your name.*
ii. *Choose a name for your virtual machine.*
iii. *Enter a username.*
iv. *Add a password and repeat the password.*
v. *Choose whether you want to log in automatically or require a login.*
vi. *Choose whether to encrypt the home folder.*
vii. *Click Continue.*

Completing the Installation

The final stage is to wait for the files to finish copying and the installation to complete. When the process is complete you will be asked to reboot.

This, of course, refers to the virtual machine and not your host Windows machine.

You can reboot in several ways such as clicking the icon in the top right corner of Ubuntu and choosing to restart or by using the reset option from the VirtualBox menu.

Install Guest Additions

You will notice that if you choose to view Ubuntu in full-screen mode that it doesn't necessarily scale correctly.

To get the best experience possible you will need to install Guest Additions.

This is a simple process:

i. Just select Devices.

ii. Then choose Install Guest Additions from the menu while running the virtual machine.

iii. A terminal window will open, and commands will run. When it has completed you will need to restart the virtual machine again.

Add two screen shots of your machine at different scales also capturing your registration number in the name of the virtual machine in addition to the screen shots of intermediate steps.

UBUNTU IS NOW GOOD TO GO.

Exercise : In your report write down the answers for the following Questions.

1. Give a brief introduction to VirtualBox
2. Advantages and disadvantage of virtual installation
3. Advantages and disadvantage of multi-booting